

WEEKS 5 – 6

Mesh Networking & Off-Grid Communication

LoRa · Meshtastic · MQTT · Resilient Networks



Session 1

LoRa · Meshtastic · Topology



Session 2

MQTT · Bridging · Resilience

Session 1

LoRa Radio Fundamentals & Meshtastic Setup

- LoRa radio fundamentals
- Meshtastic setup & configuration
- Mesh network topology

What is LoRa?

Long Range

Signals can travel 2–15 km in cities up to 40-400 km in open terrain, penetrating walls and terrain obstacles.

Low Power

Devices run for months or years on AA batteries — ideal for remote deployments.

Low Bandwidth

Not for video or voice. Think short text, sensor data, coordinates — ~250 bytes at a time.

License-free bands

868 MHz (EU) / 915 MHz (US) — no ham license required for basic use.

LoRa: Origin & History

The technology behind every Meshtastic node — and where it came from.

2009

Cycleo founded

A small startup in Grenoble, France develops chirp spread spectrum modulation for low-power radio. The core LoRa IP is born.

2012

Semtech acquires Cycleo

Semtech — a US semiconductor company — buys Cycleo and takes ownership of the LoRa patent. They begin producing the SX127x chip series used in hardware today.

2015

LoRa Alliance & LoRaWAN standardised

Semtech co-founds the LoRa Alliance with IBM, Cisco, and others. They standardise LoRaWAN — a centralised IoT protocol built on top of LoRa radio.

2018

Meshtastic project begins

An open-source project repurposes LoRa for decentralised peer-to-peer mesh networking — no gateways, no cloud, no infrastructure required.

LoRa = the radio modulation (the chirp), owned by Semtech. **LoRaWAN** = centralised IoT protocol on top of LoRa. **Meshtastic** = open, decentralised mesh — same radio, different vision.

LoRa vs. WiFi

Both are wireless — but built for completely different jobs.

	LoRa	WiFi
Range	2 – 400 km	20 – 100 m
Power	Very low (years on battery)	High (hours on battery)
Speed	0.3 – 50 kbps	10 – 10,000 Mbps
Good for	Short texts, GPS, sensors	Video, web, files
Needs router?	No — nodes talk directly	Yes — needs infrastructure
License	None (ISM band)	None (ISM band)

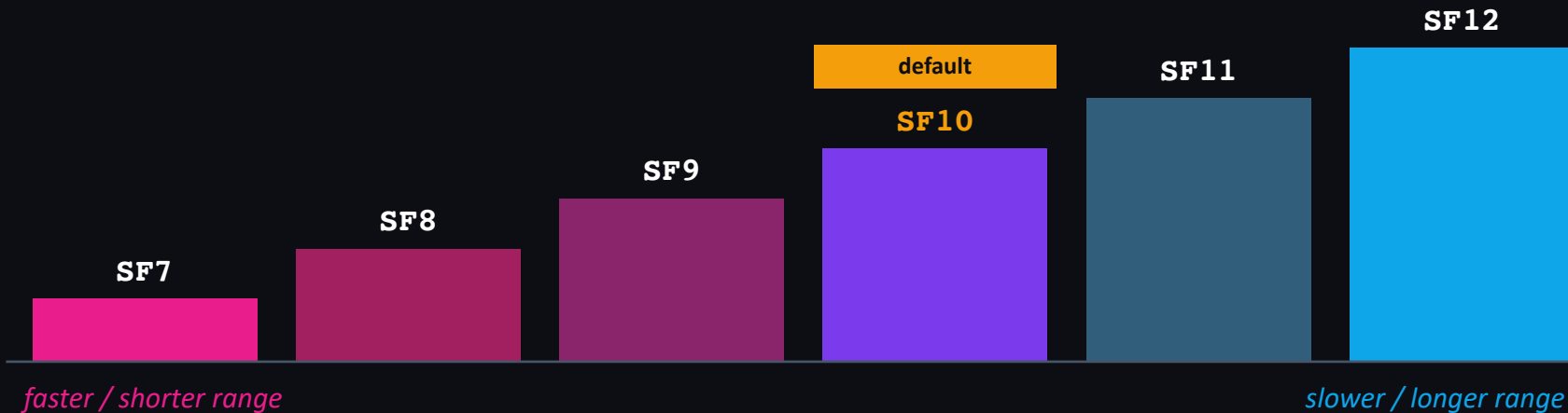
LoRa trades speed for distance and battery life. WiFi trades range for bandwidth. Neither is better — they solve different problems.

The Spreading Factor

LoRa doesn't have a fixed range. You dial the range in with a tradeoff on time.

Think of it like shouting vs. whispering slowly.

You can shout one word fast and hope your friend hears it — or you can draw each syllable out slowly and clearly so they catch every sound even from far away. A higher spreading factor is drawing it out. The message takes longer, but it carries further and survives more noise.



each step up =
2× the airtime

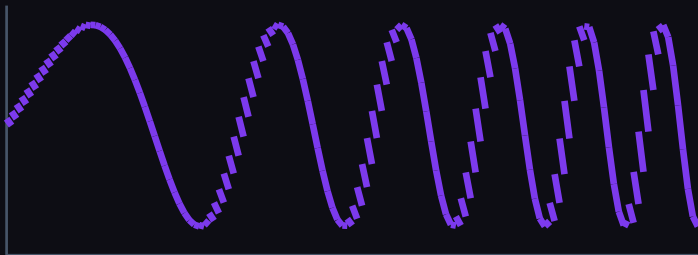
each step up =
2× the range

SF10 default =
sweet spot for city-scale mesh

The Chirp: How LoRa Encodes Data

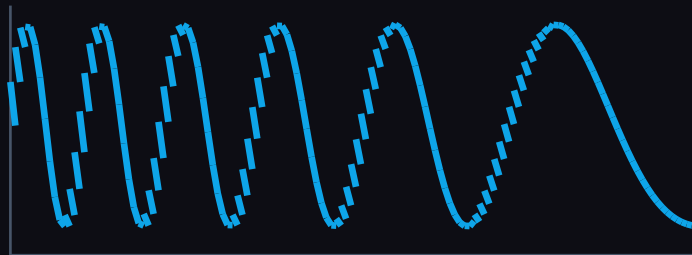
LoRa does not transmit at one fixed frequency — it sweeps continuously across a bandwidth. That sweep is a chirp.

UPCHIRP

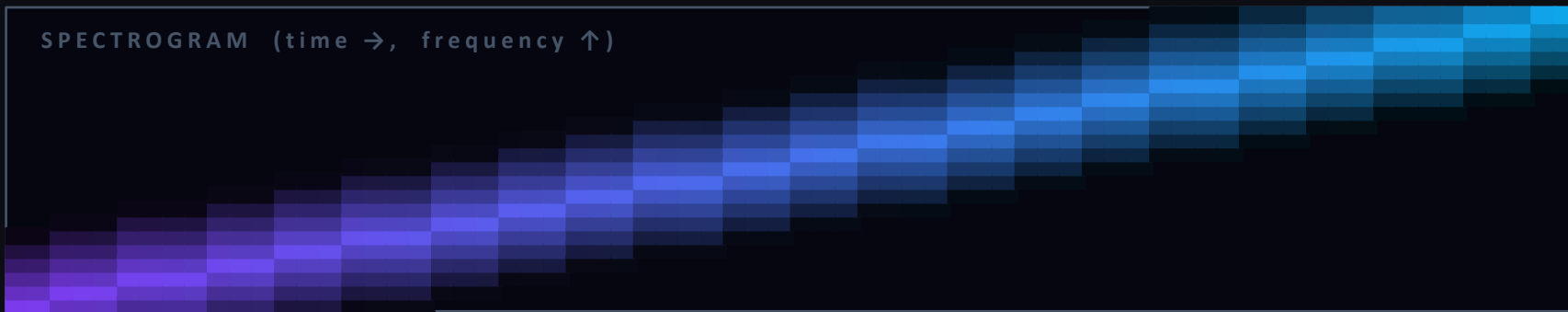


frequency sweeps low to high — encodes one symbol

DOWNCHIRP



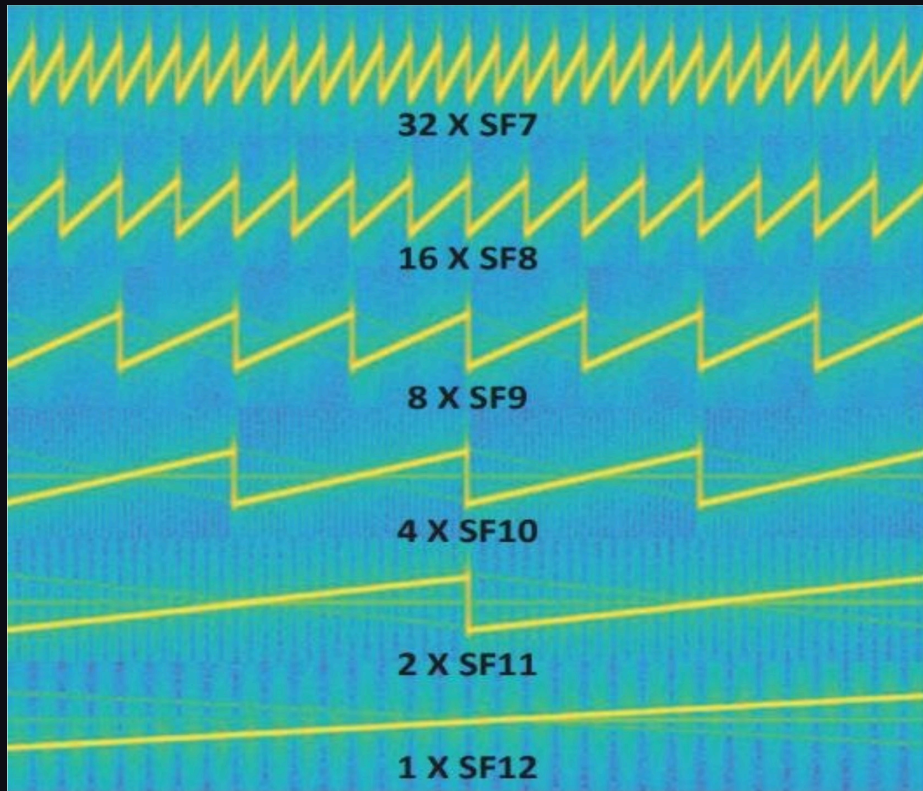
preamble — sweeps high to low, signals packet incoming



The diagonal stripe is the chirp — the receiver locks onto it even buried in noise

What Chirps Actually Look Like

Real SDR waterfall capture — each row is one spreading factor, shown at the same scale.



SF7

32 chirps visible — tiny, fast zigzags

SF8

16 chirps — slightly wider sweeps

SF9

8 chirps — getting visibly slower

SF10

4 chirps — Meshtastic default, clear diagonal

SF11

2 chirps — half the screen per sweep

SF12

1 chirp — spans the full capture width

[Watch: LoRa chirp explained — youtu.be/dxYY097QNs0](https://youtu.be/dxYY097QNs0)

What is Meshtastic?

An open-source, off-grid mesh communication platform built on LoRa radio hardware.



Works without internet, cell, or WiFi



End-to-end encrypted by default



Designed for days of battery life



GPS position sharing built-in



Phone app via Bluetooth (iOS & Android)



Python API for scripting & hacking

Common Hardware

Heltec WiFi LoRa 32

Built-in OLED screen

T-Beam / T-Echo

GPS + LoRa combo

RAK WisBlock

Modular, pro-grade

Seeed SenseCAP T1000

Ultra-compact

DIY ESP32 + SX1276

Breadboard it yourself

Setting Up Your Node

1

Flash firmware

Download Meshtastic flasher → connect USB → select your board → flash

2

Connect via Bluetooth

Open Meshtastic app (iOS or Android) → scan → pair with your node

3

Set your region

Settings → LoRa Config → Region: EU_868 (Berlin) or US_915

4

Set a channel

Default channel = 'LongFast'. Custom channels use a pre-shared key for privacy.

5

Configure your node info

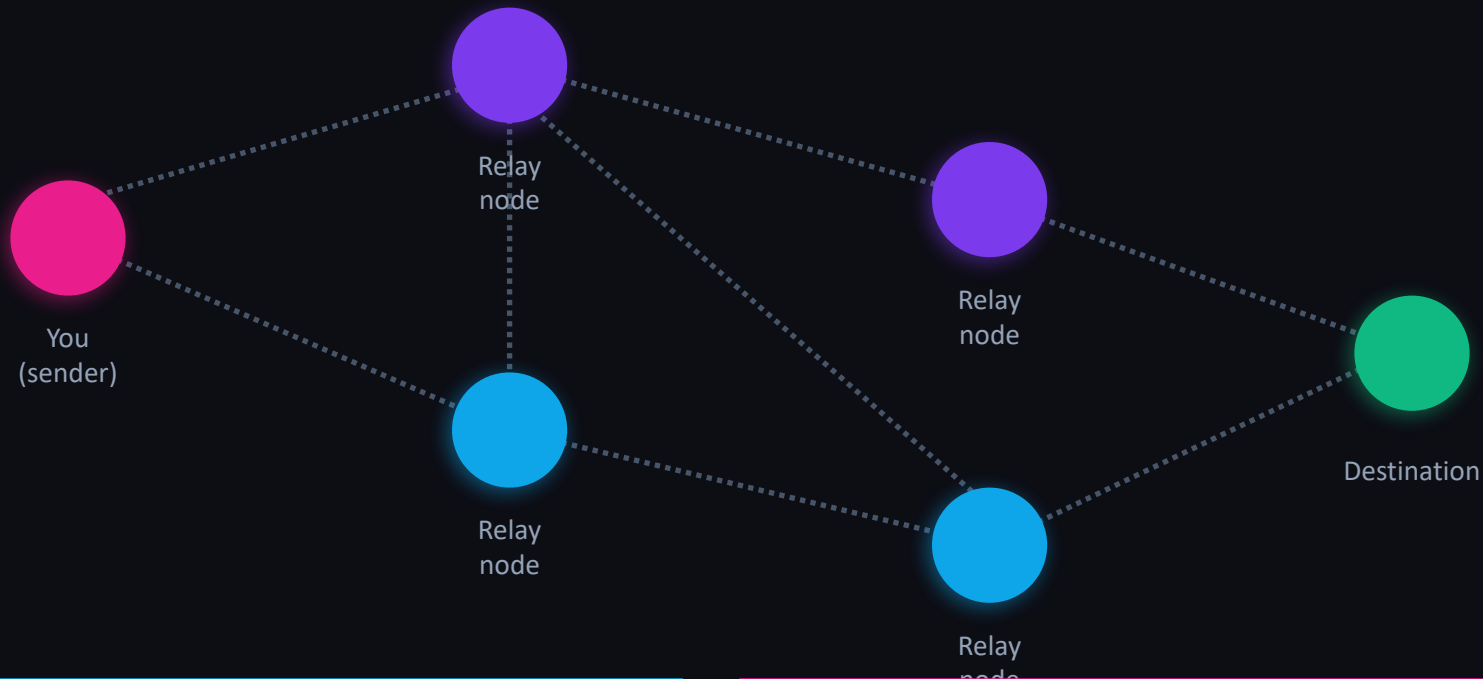
Set a short name (4 chars) + long name. This is your identity on the mesh.

6

Send a test message

Type in the app → your node broadcasts → neighbors relay it across the mesh

How a Mesh Network Thinks



Flood routing: every node rebroadcasts messages it hasn't seen before. Hops limited by TTL (default: 3).

No message ever travels one path — the mesh finds all possible routes simultaneously.

Node Roles & Channels

NODE ROLES

CLIENT

Default. Sends messages, routes for neighbors, low power mode OK.

CLIENT_MUTE

Sends & receives but does NOT relay others' messages.

ROUTER

Optimized for high-traffic relay. Always awake, high power.

REPEATER

Only forwards, never generates messages. For fixed infrastructure.

CHANNELS

Primary channel

Where all devices must agree (name + PSK). LongFast is the default.

Secondary channels

Up to 7 additional channels per node. Each has its own PSK.

Admin channel

Remote management. Used to configure nodes without physical access.

Session 2

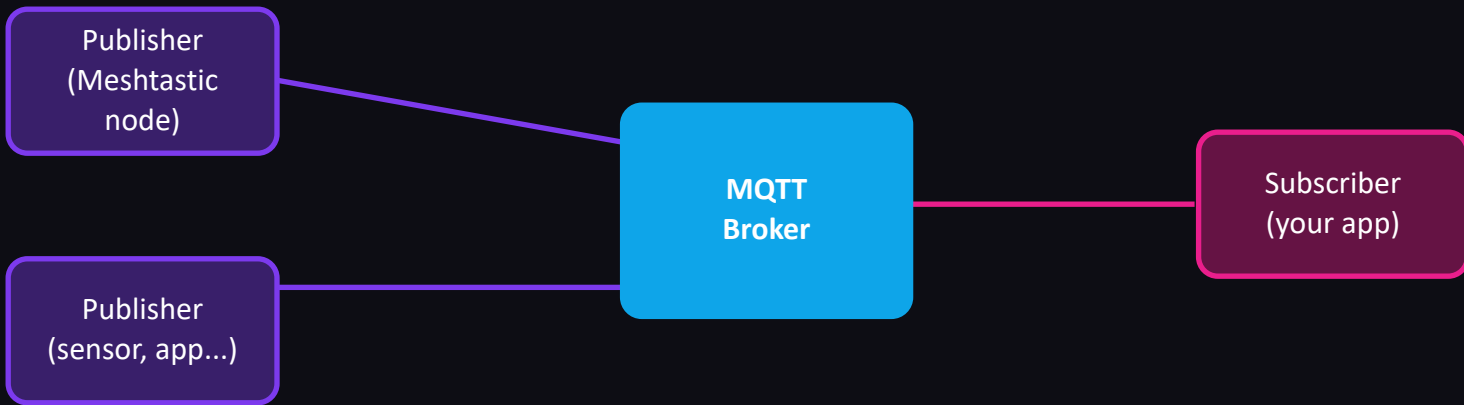
MQTT, Bridging & Designing for Resilience

- MQTT protocol
- Bridging mesh to your web server
- Designing resilient systems

What is MQTT?

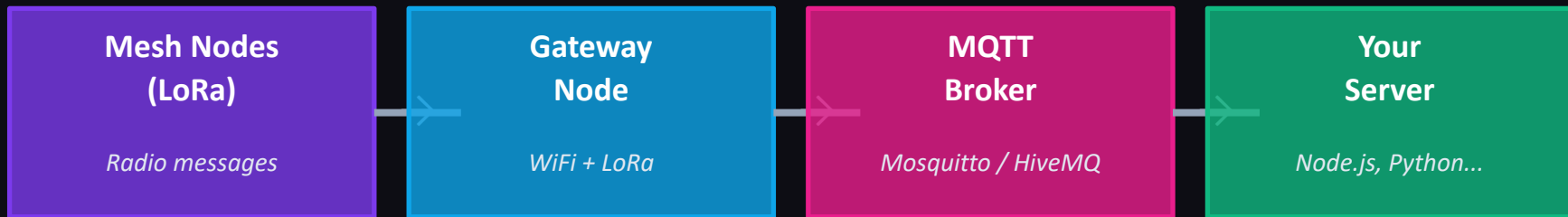
Message Queuing Telemetry Transport

A lightweight publish/subscribe messaging protocol. Think of it as a post office for machine messages — designed for low-bandwidth, unreliable networks.



Topic example: `msh/EU_868/2/json/+` — nodes publish to topics; subscribers filter by topic pattern

Bridging Mesh → MQTT → Web



CONFIGURATION STEPS

- 1 On your Meshtastic node: Settings → MQTT → enable, set broker IP, port 1883, set topic prefix
- 2 Install Mosquitto (or use a hosted broker). Test with: `mosquitto_sub -h localhost -t 'msh/#' -v`
- 3 In your server (Python/Node): subscribe to the MQTT topic, parse the protobuf JSON payload
- 4 A gateway node needs WiFi access — it bridges the LoRa mesh to your IP network

Designing for Resilience

What does it mean to build a network that survives failure?

No single point of failure

If one node dies, the network reroutes. Place nodes strategically to maximize redundancy.

Assume disconnection

Design for store-and-forward. Messages should queue and send when a path opens.

Power independence

Solar charging + LiPo = indefinite uptime. A relay node in a window runs forever.

Out-of-band management

Admin channels let you configure nodes remotely, even when you can't physically reach them.

Know your terrain

Height matters enormously. A node at 30m altitude can cover 10x more area than ground level.

Document & test degraded modes

What happens with 2 nodes? 1 node? Does your system still provide value at minimum viable scale?

Who Uses This & Why It Matters

Disaster & Emergency

- Puerto Rico after Hurricane Maria
- Turkey/Syria earthquake response 2023
- Wildfire evacuation coordination

Activism & Protest

- Hong Kong protests (Bridgefy)
- Surveillance-free coordination
- Off-grid communication in repressive contexts

Remote & Rural

- Hiking trail waypoint sharing
- Farm sensor networks
- Mountain rescue coordination

Art & Research

- Distributed media installations
- Urban sensing & data art
- Community infrastructure projects

Hands-On Exercise

Work in pairs — build a 2-node mini-mesh and bridge it to a local MQTT broker

Phase 1

20 min

Flash & configure nodes

- Flash both devices with Meshtastic firmware
- Set region to EU_868, channel to LongFast
- Give each node a unique name

Phase 2

20 min

Send messages + observe routing

- Send text between nodes via the app
- Check the hop count on received messages
- Walk around — observe signal range

Phase 3

20 min

Bridge to MQTT

- Enable MQTT on one node (set it as gateway)
- Subscribe to the topic on a local broker
- Display received messages in the terminal

Key Takeaways



LoRa = long range, low power, low bandwidth radio — perfect for off-grid text & sensor data.



Meshtastic turns cheap hardware into a self-healing, encrypted mesh network with no infrastructure needed.



Flood routing means there's no master, no central point of failure — the network thinks collectively.



MQTT bridges your mesh to the internet. One WiFi-connected node is all you need as a gateway.



Resilient design means planning for failure: redundancy, solar power, height, and degraded-mode testing.



This is artistic territory. Invisible infrastructure made visible. Networks as a medium.